25901233 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECT204 - ROBOTICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which of the following is an example of a sensor? [U] / [CO1]

   a). Motor b). Gear

   c). Camera d). Bolt

2. Explain degrees of freedom (DOF). [U] / [CO2]

   a). Total robot weight b). Speed of joints

   c). Number of ways robot can move or position d). Power consumption

3. Define On-Off control system. [K] / [CO3]

   a). Continuous proportional output b). Binary full-on or full-off based on setpoint

   c). PID tuned d). Adaptive learning

4. Select a key educational robot application. [K] / [CO4]

   a). Teaching programming b). Welding parts

c). Packing goods d). Spot inspection

5. Explain a key benefit of self-healing robots in extreme environments. [U] / [CO2]

a). Increased weight b). Minimal maintenance and resilience to hazards

c). Slower operation d). Higher costs

6. Intelligent robots are capable of: [K] / [CO1]

a). Learning and decision-making b). Only repeating tasks

c). Operating without power d). Manual-only control

7. Distinguish trajectory planning from tracking. [U] / [CO2]

a). Planning: path generation; Tracking: follow via control b). Same process

c). Planning ignores sensors d). Tracking designs path

8. Identify ROS strength. [A] / [CO3]

a). Closed source b). Ecosystem packages community

c). No Linux support d). Single robot only

9. Use of Robots in packing improves? [A] / [CO4]

a). Noise level b). Cost of materials

c). Robot size d). Speed and accuracy

10. Define the primary function of industrial robots. [K] / [CO2]

a). Mimic human movements b). Perform repetitive manufacturing tasks

c). Navigate unstructured environments d). Provide companionship

11. The main components of an ultrasonic sensor include: [K] / [CO3]

a). Battery and motor b). Switch and wire

c). Display and keyboard d). Transmitter and receiver

12. Distinguish linear vs DC motor. [U] / [CO2]

a). Linear: direct thrust no rotary; DC: continuous rotation

b). Both step-based

c). Linear low speed

d). DC linear motion

13. Environmental sensors are mainly used to measure: [K] / [CO3]

a). Mechanical motion

b). Environmental conditions

c). Electrical resistance only

d). Digital signals

14. Describe SCARA robot workspace suitability. [K] / [CO2]

a). Complex 3D assembly

b). Planar pick-place tasks

c). Heavy lifting

d). Humanoid motion

15. Identify middleware advantage. [A] / [CO3]

a). Increases complexity

b). Hardware abstraction portability

c). Removes libraries

d). Single computer only

16. Use of testing is for. [A] / [CO3]

a). Initial ideas

b). Buy materials

c). Stress safety and integration checks

d). Market analysis

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. Define robot classification. [U] / [CO1]

18. Explain the concept of path planning in robotics. [U] / [CO2]

19. Write about conceptual design steps to design a simple mobile robot for material transport. [A] / [CO3]

20. Illustrate the role of sensors and control systems in ensuring safe human–robot interaction. [An] / [CO4]

21. Define robotic locomotion. [K] / [CO2]

22. Write the basic principles of motion to a robotic arm movement example. [A] / [CO2]

23. Write the concept of feedback control to maintain the speed of a robot motor. [A] / [CO3]

24. Why are humidity sensors important in weather monitoring? Illustrate with examples [An] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. Discuss the role of robotics in modern society and its impact on daily life. [K] / [CO1]

26. Illustrate tracked and humanoid robots, including their advantages, limitations, and applications. [A] / [CO2]

27. Discuss how sensors improve the intelligence and autonomy of modern robots. [U] / [CO3]

28. Compare and analyze space robots and underwater robots in terms of environment, design requirements, sensors, and operational challenges. [An] / [CO4]

[12x2 = 24]